Unit 9, Lesson 9: Solving Quadratics Using the Quadratic Formula

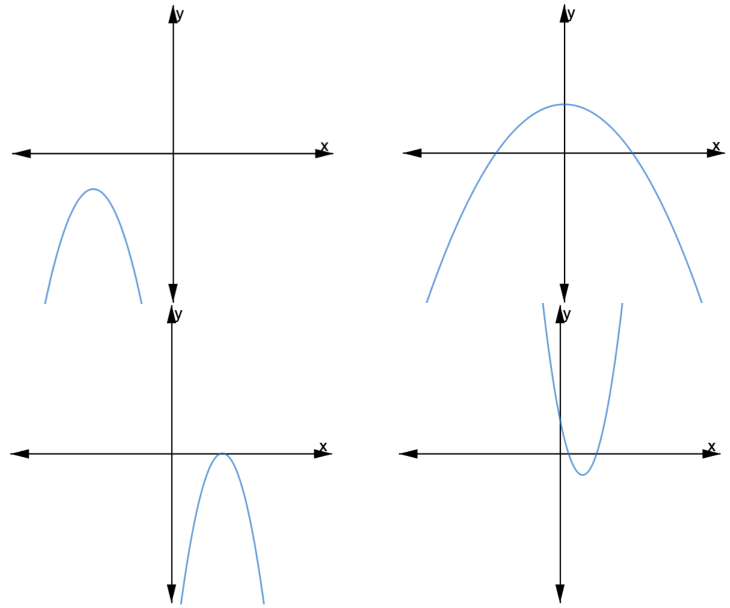
Benchmark

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

Learning Target

☐ I can solve a quadratic using the quadratic formula.

9.9.1 Warm-Up: Which One Doesn't Belong?



1. Four different parabolas are shown. Choose one graph that does not belong and explain your choice using key features.

9.9.2 Guided Instruction: Quadratic Formula

1. Rewrite $x = -5 \pm \sqrt{81}$ as two separate values.

2. Determine the two different values $x = -b \pm \sqrt{b^2 - 4ac}$ given $a = -2$, $b = -15$, and $c = -6$. Leave in square root form.

3. Determine the two different values $x = -b \pm \sqrt{b^2 - 4ac}$ given $a = 1$, $b = 4$, and $c = 8$. Leave in square root form.

The square root of a negative number is a **non-real value**.

Solving quadratics when given standard form is similar to isolating the variable x from standard form, except that standard form is set equal to 0.

Standard Form Set Equal to 0	$ax^2 + bx + c = 0$
From standard form the vertex of the parabola can be found using $\left(\frac{-b}{2a}, \frac{c}{a} - \frac{b^2}{4a^2}\right)$. Rewrite standard form to vertex form using this vertex.	$a(x-h)^2 + k = 0$ $\left(x + \frac{b}{2a}\right)^2 + \frac{c}{a} - \frac{b^2}{4a^2} = 0$
Subtract $\frac{c}{a}$ from both sides.	
Add the opposite of $-\frac{b^2}{4a^2}$ to both sides.	
In this step, a mathematician provided an equivalent form. The algebra that the mathematician used will be learned in Algebra 2. All of the steps the mathematician used are not shown.	$\left(x + \frac{b}{2a}\right)^2 = -\frac{c}{a} + \frac{b^2}{4a^2}$ $\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$

Take the square root of both sides. When taking the square root of both sides include $\pm$ to represent both of the possible outputs.	$\left(x+\frac{b}{2a}\right)^2=\frac{b^2-4ac}{4a^2}$
Subtract $\frac{b}{2a}$ from both sides.	
In this step, a mathematician provided an equivalent form. The algebra that the mathematician used will be learned in Algebra 2. All of the steps the mathematician used are not shown.	$x=\frac{-b}{2a}\pm\sqrt{\frac{b^2-4ac}{4a^2}}$ $x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}$

By isolating the variable x, a formula for solving any quadratic in standard form that is set equal to 0 was created. This formula is called the quadratic formula.

Quadratic Formula

$$\boldsymbol{x=\frac{-b\pm\sqrt{b^2-4ac}}{2a}}$$

The quadratic formula is used to solve a quadratic in standard form that is set equal to 0, $ax^2+bx+c=0$.

4. Complete the table.

Quadratic Equation	Quadratic Formula
$12x^2+8x-15=0$ $a=$ ______ $b=$ ______ $c=$ ______	**Solutions:**__________
$x^2+6x+7=-6$ x^2+6x+ _____ $=0$ $a=$ ______ $b=$ ______ $c=$ ______	**Solutions:**__________
$-5x^2-5x+18=4x+17$ $-5x^2$ ___ ___$x+$ ___ ___ $=0$ $a=$ ______ $b=$ ______ $c=$ ______	**Solutions:**__________

9.9.3 Try It: Solving With the Quadratic Formula

1. Use the quadratic formula to solve $x^2 + 4x + 4 = 0$.

2. Use the quadratic formula to solve $3x^2 - 7x - 3 = 3x + 8$.

9.9.4 Your Turn!

1. Solve $-7x^2 + 10x - 1 = 0$.

2. Determine the solution(s) for $5x^2 - 7x + 5 = 2x^2 - 17x$.

9.9.5 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered …	
One mistake I learned from today …	One thing I am not sure about … YET!
One thing I am confident I know is …	

Unit 9, Lesson 10: Applying Different Methods of Solving a Quadratic

Benchmarks

MA.912.AR.3.1 Given a mathematical or real-world context, write and solve one-variable quadratic equations over the real number system.

MA.912.AR.9.6 Given a real-world context, represent constraints as systems of linear equations or inequalities. Interpret solutions to problems as viable or non-viable options.

Learning Targets

- ☐ I can solve a quadratic using different methods.
- ☐ I can choose a method for solving a quadratic based on the context.